

CPD Activities and Reflections

Reflection 1: Project Scope Refinement and Supervisor Feedback

One important development activity during this project was the refinement of the project scope following supervisor feedback and further review of the research direction. At the early stage, the project was framed more broadly around phishing email detection and user awareness. However, as the project developed, it became clear that building another binary phishing classifier would not provide a sufficiently distinctive contribution, especially because many existing systems already focus on identifying whether an email is phishing or legitimate.

Through feedback and further planning, I refined the project into a more focused study on psychological trigger detection in suspicious emails. This changed the project from asking only whether an email is unsafe to asking how the email attempts to influence the user's judgement. The final scope therefore focused on seven psychological triggers: Urgency, Scarcity, Authority, Fear, Social Proof, Reciprocity, and Liking. This refinement helped make the project more coherent because the dataset, annotation scheme, model development, and Streamlit prototype all became aligned with the same awareness-oriented objective.

This activity improved my understanding of project scoping. I learned that a successful final year project does not need to solve every part of a cybersecurity problem. Instead, it should define a clear and manageable contribution. In this case, positioning the system as a second-stage awareness support tool made the project more realistic and defensible. It also helped me understand the importance of responding to feedback critically rather than simply adding more features. By narrowing the project scope, I was able to produce a clearer methodology, a more consistent evaluation strategy, and a prototype that directly reflects the research aim.

Reflection 2: Model Evaluation and Experimental Decision-Making

A second key development activity was the process of designing and improving the model evaluation strategy. Initially, model performance could have been judged using a single train-test split, but this would have provided only a limited view of model reliability. During the project, I adopted five-fold MultilabelStratifiedKfold

cross-validation and used Macro F1 as the primary evaluation metric. This was an important decision because the dataset was imbalanced across the seven psychological trigger labels, with Scarcity and Social Proof having far fewer positive examples than Urgency and Reciprocity.

This process helped me understand that model selection should not be based only on the highest result from a single test split. For example, SVM performed strongly as a traditional machine learning baseline, showing that TF-IDF features can capture useful lexical patterns in phishing and scam emails. However, RoBERTa achieved the strongest cross-validated Macro F1 and showed better alignment with the semantic nature of psychological trigger detection. This led to a more balanced final model selection decision, combining empirical performance with the conceptual requirements of the task.

The activity strengthened my practical machine learning skills, particularly in evaluation design, metric interpretation, and result discussion. I also learned that strong results from simpler models should not be ignored. Instead, they should be treated as meaningful baselines that help explain the task. This changed how I understood model comparison: the aim is not simply to prove that a transformer model is better, but to explain what each model reveals about the dataset and classification problem. This experience will be useful in future cybersecurity and data science work because it emphasises robust evaluation over surface-level performance claims.

Reflection 3: Prototype Development and User Awareness Design

The third major development activity was the implementation of the Streamlit prototype. This activity moved the project beyond model training and required me to consider how model outputs should be presented to non-expert users. The prototype was designed to accept suspicious email text as input, process it using the same preprocessing pipeline as the training stage, and output the predicted psychological triggers with confidence scores, explanations, and protective guidance.

Developing the prototype helped me understand the difference between technical classification output and user-facing security support. A model may produce useful predictions, but those predictions are not automatically meaningful to ordinary users. For example, simply showing that an email contains “Authority” or “Fear” is

not enough unless the user can understand what those triggers mean and why they matter. Therefore, the interface needed to translate technical predictions into practical awareness feedback, such as advising users to verify the sender independently, avoid clicking suspicious links, or treat urgent requests with caution.

This activity also highlighted the limitations of the system. Because the tool was trained only on phishing and scam emails, it should not be treated as a standalone phishing detector. Instead, it is better positioned as a second-stage explanation tool that complements an upstream phishing classifier or a user's existing suspicion. This was an important lesson in responsible system design. In cybersecurity, usability and interpretability are as important as technical accuracy because end users may not have the knowledge needed to interpret raw model outputs. Overall, the prototype development process helped me connect machine learning, human factors, and practical security awareness into a single working artefact.